

ECON 351 · Focused Practice Set

Set 9 of 10

Chapter 9

Certainty Equivalent & Risk Premium

2 questions in this practice set

Questions: Q22, Q23

Screenshots from original HW, sample questions, and mock midterm.

Question 1 of 2 (Practice Q22)

CE & RP #1 · HW2 Ex 1

Exercise 1 - Risk Aversion and Certainty Equivalent

Jackie owns shares of a company. With probability 80%, the value of the investment will go up to $x = \$200$, and with probability 20% the value of the investment will go down to $x = \$50$. Jackie's utility from income is given by the function

$$u(x) = \sqrt{x}$$

- a) Compute the expected value of the investment and Jackie's expected utility from holding this investment.
- b) Compute the certainty equivalent of this investment. In other words, the certain amount that makes Jackie indifferent between holding the investment or selling it for that certain amount (the certain amount provides the same expected utility as the investment).
- c) Compute the risk premium associated with this investment.

Question 2 of 2 (Practice Q23)

CE & RP #2 · Sample Q12

Question 12:

A consumer owns shares of a company. With probability 80% the value of the investment will go up to $x = \$144$, and with probability 20% the value of the investment will go down to $x = \$81$. The consumer's utility from income is given by the function

$$u(x) = \sqrt{x}.$$

- Compute the consumer's expected utility from holding this investment.
- Compute the expected value of this investment.
- Compute the certainty equivalent of this investment.
- Compute the risk premium associated with this investment.

Question 13:

Cuban is an investor who used all his income to buy a startup company called Shark Aquarium. With probability 50% the value of the company will increase to \$144. With probability 50% the value of the company will go down to \$64. Cuban's utility from income is

$$u(I) = \sqrt{I}.$$

He has no other source of income. Compute the minimum price P such that Cuban would be willing to sell all his shares of Shark Aquarium.

- $P = 94$
- $P = 130$
- $P = 121$
- $P = 104$
- $P = 100$

Question 14:

Using the information from the previous question, compute the associated risk premium.

- 3
- 1
- 7
- 10
- 4

Question 15:

Tom has the following utility from income: $u(I) = \sqrt{I}$. He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house (in this case, the house value becomes \$672,400). AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Tom is willing to pay?

- 23,010
- 45,375
- 74,000
- 32,375
- 15,600